# Scientific Machine Learning (SciML)

Module 2: Spectral Neural Operators

# Topics

- Review of PINNs
- Spectral Neural Operators (SNOs)
- Types of SNOs
- Implementation of Fourier Neural Operators (FNOs)

# PINNs

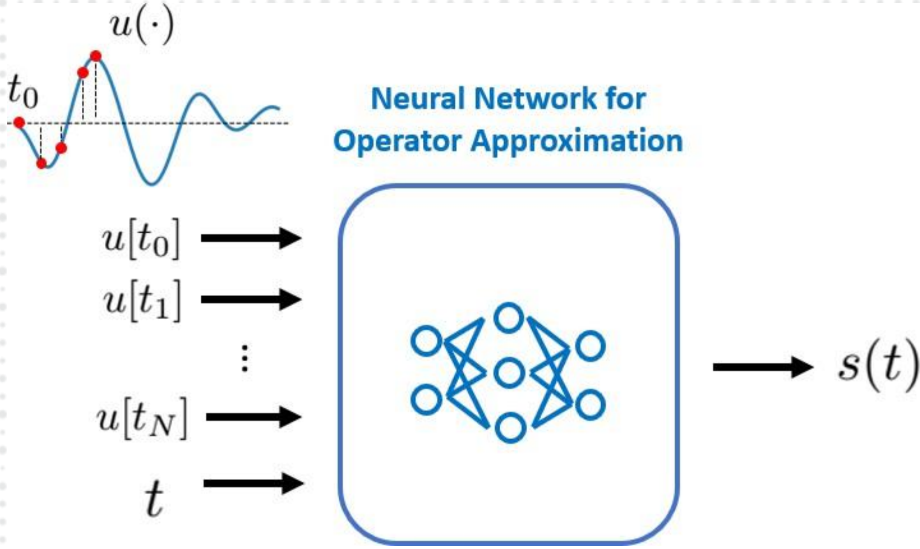
- Can be used for:
  1. Forward problems
  2. Inverse problems
  3. Equation discovery
- Purely dependent on PDEs – data is optional
- Discretize the domain using collocation points
- Build the loss using PDE, ICs, BCs and data (optional)

# Neural Operator

- Learn $G: a(x) \longmapsto u(x)$
- Supervised learning (require data)
- Input:
  - Forcing function, ICs
  - Discrete data points
- Output:
  - Steady state
  - Time dependent

# Neural Operator

- $u_0 \rightarrow u_1 \rightarrow u_2 \rightarrow \cdots \rightarrow u_{t-1} \rightarrow u_t$
- $u_{k+1}(x) = \sigma(W u_k(x) + \mathcal{K}(a; \phi) u_k)(x))$
  - $W$: Linear operator (bias vector)
  - $\mathcal{K}$: Kernel integral operator (weights)



# Spectral Neural Operator

- Transforms the input into frequency domain before learning
- PDEs exhibit simple behavior in frequency domain

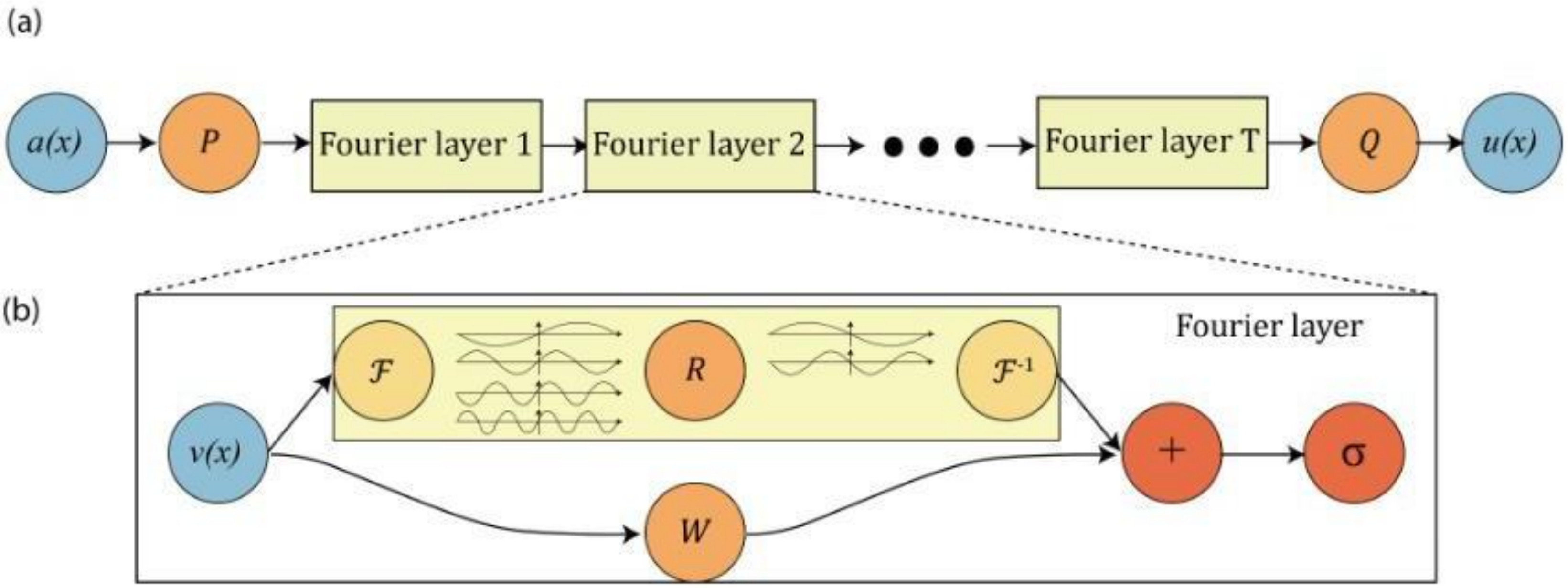
| Name | Time domain | Frequency domain |
|---|---|---|
| Wave Equation | $u_{tt} = c^2 u_{xx}$ | $-\omega^2 \hat{u} = -c^2 k^2 \hat{u}$ |
| Heat Equation | $u_t = \alpha u_{xx}$ | $i\omega \hat{u} = -\alpha k^2 \hat{u}$ |
| Burgers Equation | $u_t + u u_x = D u_{xx}$ | $\mathrm{i}\omega \hat{u} + \mathrm{i}k(\hat{u} * \hat{u}) = -D k^2 \hat{u}$ |

# Types of SNOs

| Type | Basis | Reference |
|---|---|---|
| Fourier Neural Operator (FNO) | Fourier | Li et al., *ICLR 2021* |
| Wavelet Neural Operator (WNO) | Wavelet | Tripura & Chakraborty, *Sci. Reports 2022* |
| Graph Neural Operators (GrNO) | Graph Laplacian | Li et al., *NeurIPS 2022* |

# Fourier Neural Operator

- Perform convolutions in Fourier space using FFT




Li, Zongyi, et al. "Fourier neural operator for parametric partial differential equations." *arXiv:2010.08895* (2020).

1. Lifting (P)
2. Fourier Transform $\mathcal{F}$
3. Linear transform on the lower modes: R
4. Inverse Fourier transform: $\mathcal{F}^{-1}$
5. Reconstruction (Q)

Loss: $\Sigma \frac{||\widehat{y_i} - y_i||}{||y_i||}$

# 1D Burgers' Equation

- PDE: $u_t + uu_x = u_{xx}$
- Domain: $\Omega = [0,1]; t = [0,1]$
- IC: $u(x, 0) = u_0(x)$
- Input: $a(x) = [u_0(x), X]$
- Output: $u(x, 1)$

# Live

- Complete discussion on PINNs
  - Review of the code
  - Challenges and limitations of PINNs
- FNO implementation of 1D Burgers' equation
- Challenges and limitations of FNOs

# THANK YOU